

INFO8010: Interior Design Style Transfer

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I. INTRODUCTION

Interior design visualization remains challenging for non-professionals. Most people struggle to imagine how their spaces would look in different design styles. Traditional visualization methods require either professional design skills or extensive photo editing expertise.

In this paper, we explore a machine learning approach for transforming room interior images between design styles. Our method investigates an enhanced CycleGAN architecture that aims to preserve structural elements while changing aesthetic features.

Our work examines:

- Adapting CycleGAN for room style transfer
- Implementing multi-scale discriminators and improved residual blocks
- Testing a structural preservation loss to maintain architectural elements
- Evaluating results using LPIPS and FID metrics

We focus specifically on transferring between coastal and eclectic living room styles. Our approach works with unpaired datasets, addressing the practical limitation that paired before/after images of the same room in different styles are rarely available.

II. RELATED WORK

A. Image-to-Image Translation

Image-to-image translation has seen significant progress in recent years. Pix2Pix [1] introduced a general framework for supervised image translation using conditional GANs, requiring paired training data. CycleGAN [2] extended this approach to the unpaired setting by introducing cycle consistency loss, enabling training without directly corresponding before/after images. This unpaired approach is particularly valuable for interior design applications, where obtaining before/after photos of the same room in different styles is difficult.

B. Style Transfer

Neural style transfer, explored by Gatys et al. [3], showed the ability to transfer artistic styles to photographs using feature representations from pre-trained networks. Subsequent approaches improved efficiency [4] and versatility [5]. In particular, adaptive instance normalization (AdaIN) [5] and whitening and coloring transforms (WCT) [6] demonstrated better separation of content and style. While these methods work well for artistic style transfer, they often struggle with the structural preservation needs of interior design applications.

C. Diffusion Models

More recently, diffusion-based generative models have shown exceptional capabilities in image generation and manipulation tasks. Denoising diffusion probabilistic models (DDPM) [7] introduced a new paradigm for high-quality image synthesis. These techniques have evolved into powerful text-to-image models such as DALL-E 2 [8] and latent diffusion models [9].

While these models can produce impressive style transformations with text guidance, they typically require large-scale training data and computational resources. Some recent work has explored using diffusion models specifically for image-to-image translation tasks [10], though often with different training requirements compared to GAN-based approaches.

D. Interior Design Applications

Image translation and style transfer techniques have been applied to interior design in both academic research and commercial applications. Several existing tools offer interior visualization capabilities.

We chose to work with the classical CycleGAN architecture for its established ability to work with unpaired image datasets and its accessible computational requirements compared to newer diffusion-based approaches. This makes it well-suited for interior design applications where paired before/after images are rarely available.

III. METHODOLOGY

A. Dataset

Our project uses a curated dataset of living room interior images from the MMIS (Multi-Modal Interior Style) dataset [11]. This dataset provides a comprehensive collection of interior design images with clear style categorizations, specifically focusing on living room spaces.

The dataset characteristics include:

- **Total images:** 2,069 living room images
- **Coastal style:** 1,049 images featuring light, airy spaces with natural materials, neutral colors, and beach-inspired elements
- **Eclectic style:** 1,020 images showcasing bold, mixed patterns, vibrant colors, and diverse decorative elements from various design periods
- **Image quality:** Professional interior photography with consistent lighting and composition
- **Organization:** Images directly organized into style-specific directories (`dataset/coastal/` and `dataset/eclectic/`)

We chose **coastal** and **eclectic** styles due to their distinct visual characteristics. Coastal living rooms typically feature light wood tones, white and blue color palettes, natural textures, and minimalist furniture arrangements. In contrast, eclectic living rooms combine bold colors, mixed patterns, diverse furniture styles, and rich textures. This strong stylistic contrast ensures that style transfer effects are pronounced and easily evaluable.

For training our model, we split each style category into train (70%), validation (15%), and test (15%) sets. This resulted in approximately 734 coastal and 714 eclectic living room images for training, with proportional numbers for validation and testing.

B. Model Architecture

Our architecture is based on the classical CycleGAN [2] framework. The overall system consists of two generator networks and two discriminator networks, enabling bidirectional translation between coastal and eclectic living room styles. Figure 1 illustrates the high-level architecture of our system.

1. Generator Architecture

Our generator network uses a ResNet-based architecture for living room style transfer:

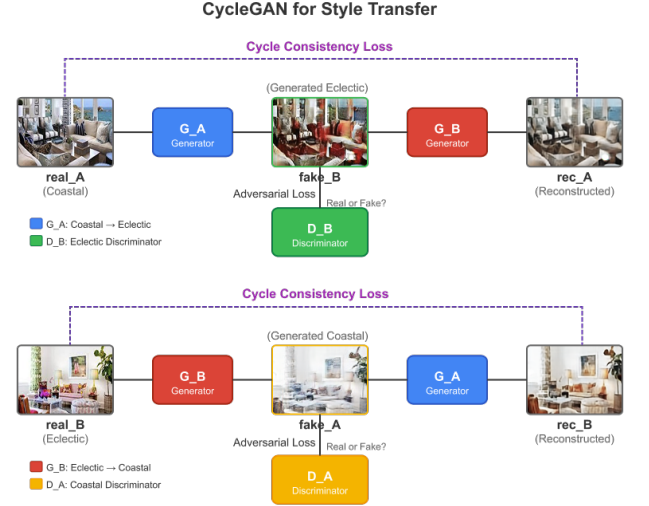


FIG. 1: CycleGAN architecture for living room style transfer between Coastal and Eclectic styles, showing generators (G_A , G_B), discriminators (D_A , D_B), and cycle consistency loss.

- Initial reflection padding and 7×7 convolution to preserve edge information
- Two downsampling layers with stride-2 convolutions and instance normalization
- Nine residual blocks for feature transformation while maintaining spatial information
- Bilinear interpolation upsampling followed by 3×3 convolutions to reduce checkerboard artifacts
- Final reflection padding and 7×7 convolution with Tanh activation

The dual generator structure allows us to create both Coastal $\rightarrow$ Eclectic (G_A) and Eclectic $\rightarrow$ Coastal (G_B) transformations. This bidirectionality enables cycle consistency training without requiring paired images.

2. Multi-scale Discriminator

We implemented a PatchGAN discriminator architecture that operates on image patches rather than the entire image:

- PatchGAN architecture that classifies overlapping patches as real or fake
- Spectral [12] normalization for training stability
- Focuses on local texture and style features rather than global image structure

This approach helps the model distinguish between coastal and eclectic styles by focusing on local style characteristics like textures, patterns, and color combinations typical of each design style.

3. Perceptual Network

C. Loss Functions

Our training objective combines two main loss terms following the classical CycleGAN approach:

$$\mathcal{L} = \mathcal{L}_{GAN} + \lambda_{cycle}\mathcal{L}_{cycle} \quad (1)$$

1. Adversarial Loss

We use the GAN loss with mean squared error (MSE) for training stability:

$$\mathcal{L}_{GAN}(G, D) = \mathbb{E}_y[(D(y) - 1)^2] + \mathbb{E}_x[D(G(x))^2] \quad (2)$$

2. Cycle Consistency Loss

This loss ensures that transformed images can be mapped back to their original versions:

$$\mathcal{L}_{cycle} = \mathbb{E}_x[||G_B(G_A(x)) - x||_1] + \mathbb{E}_y[||G_A(G_B(y)) - y||_1] \quad (3)$$

This classical CycleGAN formulation allows the generators to perform substantial style transformations while maintaining cycle consistency, which is essential for unpaired training scenarios where direct supervision is unavailable.

IV. IMPLEMENTATION

A. System Overview

We implemented our system in Python using PyTorch. The system handles the complete pipeline from dataset preparation to inference, with special attention to preserving architectural elements during style transformation.

B. Project Structure

Our codebase is organized into several modular components:

- **config.py:** Configuration file containing all hyperparameters, including batch size, learning rates, model dimensions, and loss weights. This configuration file contains the standard parameters which

can then be modified by specifying the desired values in the command line.

- **data_preparation.py:** Handles dataset operations including loading bathroom images, splitting into train/validation/test sets (70%/15%/15%), and creating appropriate data loaders with transformations for resizing, cropping, and normalization.
- **train.py:** Implements the training loop with both generator and discriminator optimization steps, periodical checkpoint saving, and sample visualization. Integrated with Weights & Biases to monitor the progress.
- **evaluate.py:** Computes quantitative metrics including LPIPS for measuring perceptual similarity and FID for assessing generation quality. Creates visualizations for qualitative assessment.
- **inference.py:** Applies the trained model to new bathroom images.
- **models/:** Contains the neural network architecture implementation:
 - **cycle_gan.py:** Implements the core CycleGAN model class, coordinating the interaction between generators, discriminators, and multiple loss functions.
 - **networks.py:** Defines the ResNet generator, multi-scale discriminator, and perceptual network architectures.
 - **losses.py:** Implements the loss functions including GAN, cycle consistency, identity, and structural preservation losses.
- **utils.py:** Provides utility functions for image processing, tensor-to-image conversion, and visualization.

C. Training Process

We trained our model using the following hyperparameters:

- **Batch size:** 8, providing stable gradient estimates while accommodating memory constraints.
- **Learning rates:** Asymmetric learning rates with generator at 0.0002 and discriminator at 0.0001 using Adam optimizer ($\beta_1 = 0.5$, $\beta_2 = 0.999$). The lower discriminator learning rate prevents it from becoming too dominant during training.
- **Training epochs:** 500, providing sufficient iterations for the model to learn meaningful style transfer between coastal and eclectic designs.

- **Image size:** 256×256 , offering a good balance between preserving room details while keeping reasonable computational requirements.
- **Loss weights:** $\lambda_{cycle} = 5.0$. We use only cycle consistency loss alongside the adversarial loss, following a simplified classical CycleGAN approach that allows more creative freedom in style transformation.

The training process included several strategies for stability:

- **Asymmetric learning rate scheduling:** We implemented different scheduling strategies for generator and discriminator to address the common problem of discriminator dominance in GAN training. The generator uses CosineAnnealingWarmRestarts ($T_0=50$, $T_{mult}=2$) to provide periodic learning rate boosts that help escape local minima. The discriminator uses CosineAnnealingLR with steady decay to prevent it from becoming too powerful relative to the generator.
- **Image buffer of size 50** for discriminator training to reduce oscillation by using a history of generated images
- **Spectral normalization** in the discriminator to enforce Lipschitz continuity and improve training dynamics
- **Reflection padding** to reduce border artifacts

Weights & Biases allowed us to observe loss curves for adversarial and cycle consistency components, learning rate schedules, and generated samples throughout training.

D. Training Monitoring

We integrated Weights & Biases (wandb) into our training pipeline to track model performance throughout the 100 epochs. By monitoring various loss components individually, we gained valuable insights into the model’s learning dynamics and convergence patterns.

1. Overall Loss Progression

The generator loss (Figure 7) shows clear correlation with the learning rate restart schedule. The loss exhibits periodic spikes corresponding to learning rate restarts at epochs 50, 150, and 350, followed by gradual improvement. Notably, after the final warm restart at epoch 350, the generator loss shows a sustained decreasing trend from approximately 2.8 to 1.8, indicating significant learning progress in the final training phase.

The discriminator loss (Figure 8) demonstrates steady improvement throughout training, decreasing from initial values around 0.5 to approximately 0.3 by epoch 500. The discriminator shows stable learning without the periodic disruptions seen in the generator, consistent with its steady learning rate decay schedule.

The validation losses (Figures 9 and 10) reveal interesting training dynamics. While the validation generator loss shows fluctuations throughout training, a positive trend emerges after the final warm restart at epoch 350, where the generator shows improved performance. Conversely, the validation discriminator loss shows a concerning upward trend in later epochs, rising from approximately 0.25 to 0.35 by epoch 500. This divergence suggests that after the final restart, the generator began learning more effectively while the discriminator’s validation performance degraded. The high validation losses indicate potential overfitting and suggest that longer training with proper regularization could yield better generalization results.

2. Learning Rate Schedule

The generator and discriminator learning rates (Figures 11 and 12) show our asymmetric scheduling approach. The generator learning rate follows CosineAnnealingWarmRestarts with clear periodic restarts at epochs 50, 150, and 350, providing learning rate boosts that help maintain generative diversity. The discriminator learning rate follows a steady exponential decay from 0.0001 to near zero, preventing discriminator dominance while maintaining training stability.

3. Domain-Specific GAN and Discriminator Losses

The domain-specific GAN losses (Figures 13 and 14) show distinct behaviors for each domain. Domain A exhibits an upward trend after epoch 300, reaching values around 0.55 by the end of training, indicating increased difficulty in fooling the discriminator. Domain B shows more stable performance with gradual improvement until epoch 400, followed by slight deterioration.

The discriminator losses for both domains (Figures 15 and 16) demonstrate effective learning. Domain A discriminator loss decreases steadily from 0.25 to approximately 0.15, while domain B shows similar convergence from 0.25 to 0.18. Both discriminators maintain stable performance in later epochs, suggesting good adversarial balance.

4. Cycle Consistency

The cycle consistency losses (Figures 17 and 18) demonstrate effective learning of the bidirectional mapping. Both domains show strong convergence, starting from values around 1.3 and decreasing to approximately 0.5-0.6 by training completion. The cycle losses exhibit periodic spikes corresponding to the generator learning rate restarts, but show consistent recovery and improvement. Notably, both domains achieve their best cycle consistency in the final epochs after the last warm restart, supporting our observation that the generator learned most effectively in this period.

E. Evaluation Metrics

We evaluated our model using two complementary metrics:

- **LPIPS** (Learned Perceptual Image Patch Similarity): This perceptual metric measures similarity between images based on features extracted from pre-trained networks. We used LPIPS to evaluate two aspects:
 - Cycle consistency: How well the original image is reconstructed after a full translation cycle ($A \rightarrow B \rightarrow A$)
 - Structural preservation: How well architectural elements are maintained when comparing original and style-transferred images
- **FID** (Fréchet Inception Distance): This metric measures the distance between feature distributions of real and generated images using the Inception-v3 network. Lower FID scores indicate generated images more similar to real ones, providing an assessment of overall generation quality and style transfer effectiveness.

These quantitative metrics, combined with qualitative visual assessment, allowed us to evaluate the performance of our bathroom style transfer system.

V. EXPERIMENTS AND RESULTS

We conducted an evaluation of our style transfer model, inspecting both quantitative metrics and qualitative visual results. Our evaluation focused on the bidirectional style transfer between coastal and eclectic living room styles.

A. Quantitative Evaluation

Table I presents the quantitative evaluation metrics of our style transfer model after training for 100 epochs.

Metric	Value
LPIPS Cycle Consistency A	0.2529
LPIPS Cycle Consistency B	0.2544
LPIPS Average Cycle Consistency	0.2536
LPIPS Structural A to B	0.2620
LPIPS Structural B to A	0.2163
LPIPS Average Structural	0.2391
FID A to B	186.0744
FID B to A	199.4874
FID Average	192.7809

TABLE I: Evaluation metrics for our living room style transfer model.

The LPIPS scores measure perceptual similarity (lower is better), with scores around 0.25 indicating reasonable preservation of content while allowing for style changes. Our model achieves reasonable cycle consistency, with an average score of 0.2536. This suggests that images can be transformed to the target style and back while maintaining content features. The structural preservation scores are similar, with slightly better preservation when converting from Eclectic to Coastal (0.2163) compared to Coastal to Eclectic (0.2620).

FID scores measure the distribution distance between real and generated images. The FID scores (186.0744 for A to B and 199.4874 for B to A) indicate that while the generated images capture stylistic elements of the target domain, there remains a gap between the distribution of generated images and real images. This is expected given the complexity of living room designs and the challenging nature of transferring between distinct design styles.

B. Qualitative Analysis

1. Coastal to Eclectic Transfer

Figure 3 shows examples of transforming coastal living room styles to eclectic style.

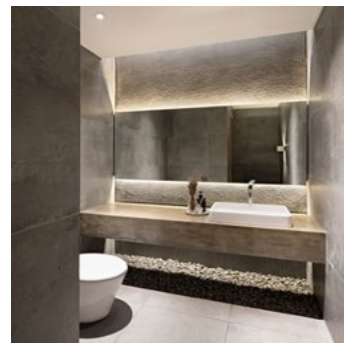


FIG. 2: Real Coastal living room (Real A)



(a) Generated Eclectic
(Fake B)

(b) Reconstructed
Coastal (Rec A)

FIG. 3: Style transfer results from Coastal to Eclectic and back

Our model transforms coastal living rooms when converting to eclectic style by introducing bolder colors and patterns. We observe that the model adds vibrant decorative elements and transforms neutral coastal palettes into more diverse color schemes typical of eclectic design. The overall room structure—furniture placement and architectural elements—is preserved during the transformation.

2. Eclectic to Coastal Transfer

Figure 5 shows the transformation from eclectic to coastal living room style.



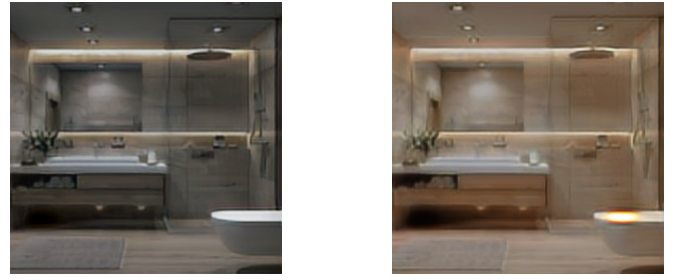
FIG. 4: Real Eclectic living room (Real B)

When converting from Eclectic to Coastal style, our model simplifies the color palette and reduces visual complexity. The model transforms bold patterns and vibrant colors into the more neutral, calming tones characteristic of coastal design. Decorative elements are toned down to create the clean, airy aesthetic typical of coastal living rooms.

C. Cycle Consistency Analysis

Figure 6 illustrates the full cycle of transformations for both directions.

The cycle consistency results show that after a full transformation cycle, images return to a state visually



(a) Generated Coastal
(Fake A)

(b) Reconstructed
Eclectic (Rec B)

FIG. 5: Style transfer results from Eclectic to Coastal and back



FIG. 6: Complete style transfer cycles. Coastal → Eclectic → Coastal. Eclectic → Coastal → Eclectic.

similar to their originals. For the Coastal → Eclectic → Coastal cycle, the reconstructed image recovers the neutral tones and clean aesthetic, while for the Eclectic → Coastal → Eclectic cycle, the model successfully retrieves the bold patterns and vibrant colors of eclectic design.

D. Structural Preservation

Our approach focuses on maintaining room structure while allowing meaningful style changes. As visible in Figures 3 and 5, our model preserves the living room layout, furniture positions, and architectural boundaries while transforming colors, patterns, and decorative elements.

The LPIPS structural scores (0.2620 for A to B and 0.2163 for B to A) support our visual observation that structural elements are largely preserved during style transfer. The better score for Eclectic to Coastal transformation suggests that simplifying bold patterns into neutral coastal aesthetics may be more straightforward than the reverse transformation.

Looking at the transformation examples, we can observe that furniture positions and room layouts remain consistent through the style transfer process, while color

palettes, patterns, and decorative elements are modified to match the target aesthetic. This demonstrates the model’s ability to distinguish between spatial content that should be preserved and stylistic elements that should be transformed.

VI. DISCUSSION

While our CycleGAN approach produces promising results, our project has several limitations that require critical examination.

A. Limitations of Style Transfer for Interior Design

Our experiments demonstrate that the current approach can effectively transform certain stylistic elements like colors, patterns, and decorative elements, but falls short in making more substantial changes that may be desired in interior design contexts. The model primarily alters the “atmosphere” of the living rooms—adding bold patterns when converting to eclectic style or simplifying to neutral tones for coastal style—but struggles with more comprehensive transformations.

The high FID scores (average of 192.78) indicate a significant distribution gap between generated and real images, suggesting that our transformations are not fully capturing the authentic characteristics of each design style. This limitation may stem from the challenging nature of style transfer between distinctly different design aesthetics.

B. Insufficient Feature Engineering

One shortcoming of our implementation is the minimal feature engineering specific to interior design. We used a classical CycleGAN approach without developing domain-specific features tailored to living room style transfer. For example:

- No semantic segmentation to distinguish between different living room elements (furniture, walls, decorative items, etc.)
- Lack of style-specific transformations that could better capture design differences
- No specialized attention mechanisms to focus on style-defining elements like patterns, textures, and color schemes
- Absence of design-specific perceptual features for more meaningful loss functions

These limitations restrict the model’s ability to make targeted, design-relevant transformations, instead producing more general stylistic changes.

C. Structural Preservation vs. Design Transformation

A significant limitation lies in the classical CycleGAN’s tendency to preserve spatial structure. While this helps maintain room layouts, interior design transformation often benefits from changes to furniture arrangements and decorative elements. The model’s inclination to preserve structure prevents it from:

- Replacing coastal-style furniture with bold eclectic pieces
- Transforming decorative elements between styles
- Changing patterns and textures in a style-appropriate manner
- Modifying lighting and accessories which are key style indicators

This reveals a fundamental limitation: successful interior style transfer often benefits from changes to decorative elements, yet the model tends to preserve existing spatial arrangements. This constraint limits the system’s ability to create more dramatic style transformations.

D. CycleGAN Limitations for Design Applications

Our results suggest that CycleGAN, even with enhancements, may not be the optimal architecture for sophisticated interior design transformations. The inherent limitations include:

- Inability to generate new elements not present in the original image
- Limited capacity to understand the semantic meaning of different room components
- Tendency to produce artifacts in complex regions with many geometric details

These architectural limitations are evident in our results, where style transfer is largely limited to color shifts and texture changes rather than meaningful design transformations.

E. Alternative Approaches

The limitations identified suggest that newer generative models, particularly diffusion models, might be better suited for interior design style transfer. Recent text-to-image and image-to-image diffusion models such as Stable Diffusion have shown superior capabilities in:

- Understanding semantic content of scenes

- Generating new objects and elements while maintaining context
- Producing high-fidelity transformations guided by text prompts
- Preserving important elements while allowing for targeted modifications

A diffusion-based approach would potentially allow for both preservation of room structure and transformation of specific design elements through controlled generation. This could enable more meaningful style transfers that actually replace industrial fixtures with Scandinavian ones, rather than merely adjusting their appearance.

Furthermore, we could also address the fundamental limitation of our current system, which applies the same transformation logic to all bathroom elements regardless of their design significance.

VII. CONCLUSION AND FUTURE WORK

In this paper, we presented a deep learning approach for living room interior design style transfer using a classical CycleGAN architecture. Our system transforms living room images between coastal and eclectic styles, changing colors, patterns, and decorative elements while preserving the room structure. The quantitative results, with LPIPS cycle consistency scores around 0.25, show that our model achieves a reasonable balance between style transformation and content preservation. Our visual results confirm this finding, showing clear stylistic changes from neutral, airy coastal aesthetics to bold, vibrant eclectic designs and vice versa.

Despite these achievements, our approach has limitations in making substantial decorative and furniture modifications necessary for complete style transformations. Future work should incorporate semantic understanding of room elements to enable more targeted transformations, potentially allowing specific decorative elements to be redesigned while maintaining overall room structure. Additionally, exploring more advanced generative models, such as diffusion-based approaches, could overcome the limitations of CycleGAN and enable more sophisticated interior design transformations with improved fidelity. In conclusion, while our current system shows the feasibility of automated room style transfer between distinct design aesthetics, the path toward truly comprehensive design visualization will require more nuanced approaches that better balance preservation and transformation.

Appendix A: Training Visualizations

This appendix contains the complete set of training visualization charts generated during our model training process. These visualizations were captured using Weights & Biases (wandb) monitoring tool over the course of 100 epochs.

1. Overall Loss Charts

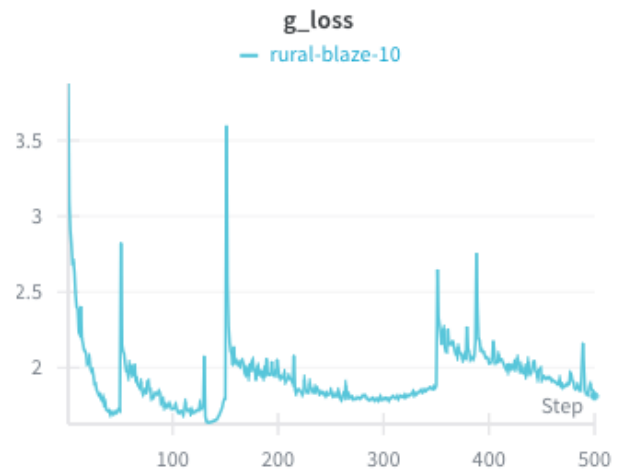


FIG. 7: Generator loss (g_loss)

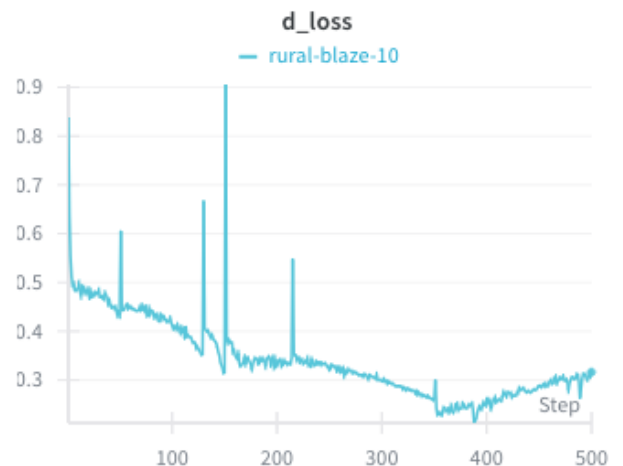


FIG. 8: Discriminator loss (d_loss)

2. Learning Rate Schedule

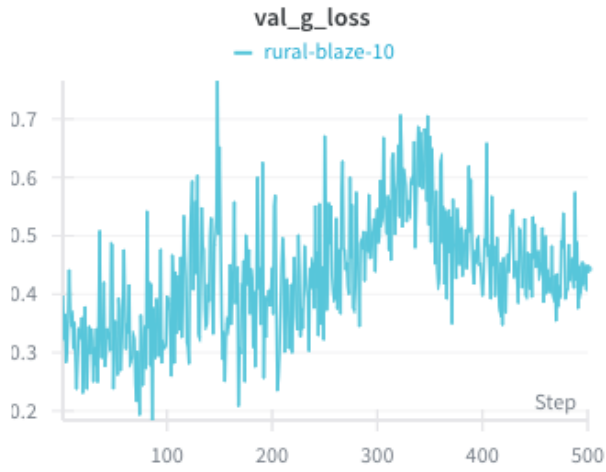


FIG. 9: Validation generator loss (val_g_loss)

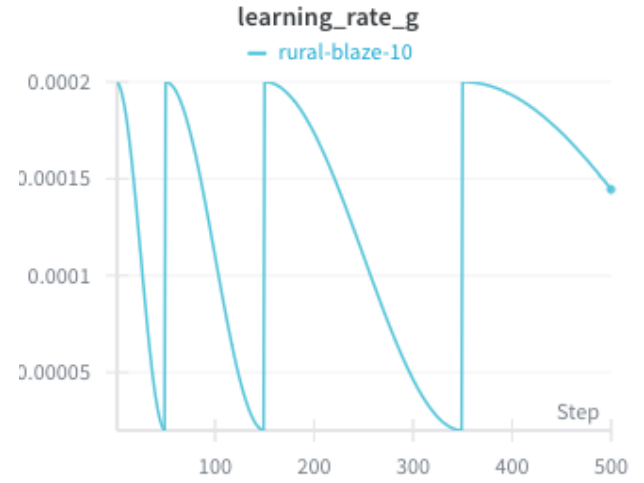


FIG. 11: Generator learning rate schedule (lr_g)

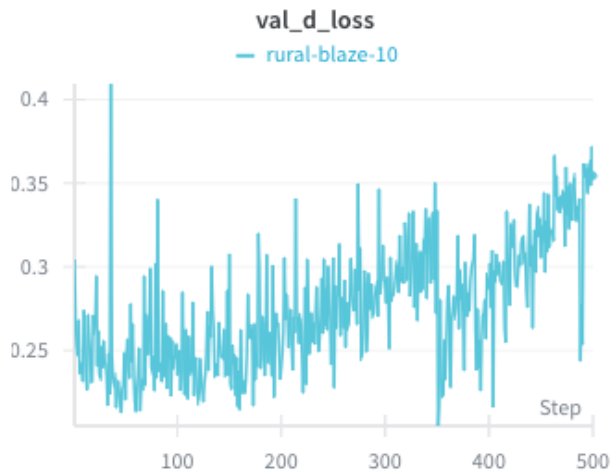


FIG. 10: Validation discriminator loss (val_d_loss)

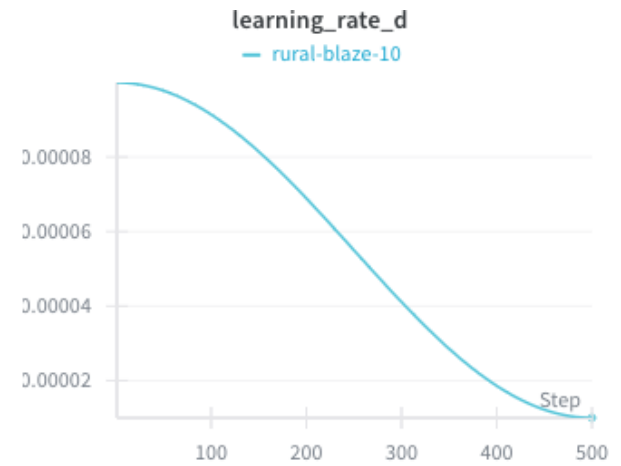


FIG. 12: Discriminator learning rate schedule (lr_d)

3. Domain-Specific Adversarial Losses

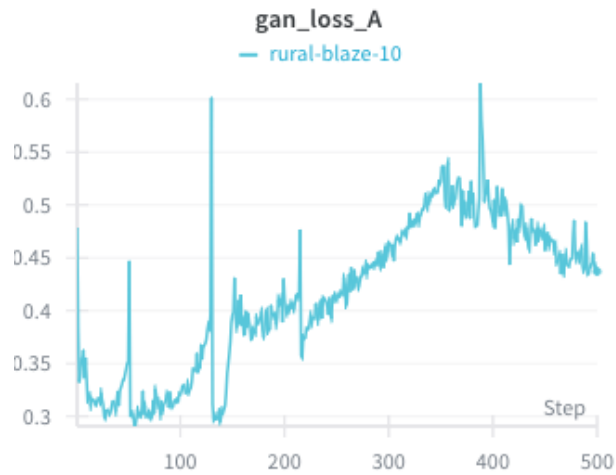


FIG. 13: GAN adversarial loss for domain A (gan_loss_A)

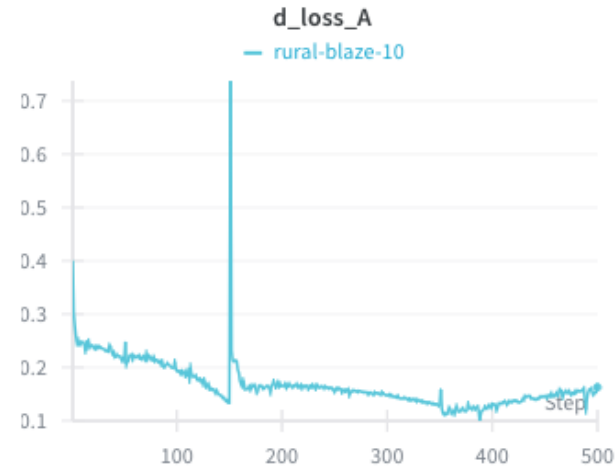


FIG. 15: Discriminator loss for domain A (d_loss_A)

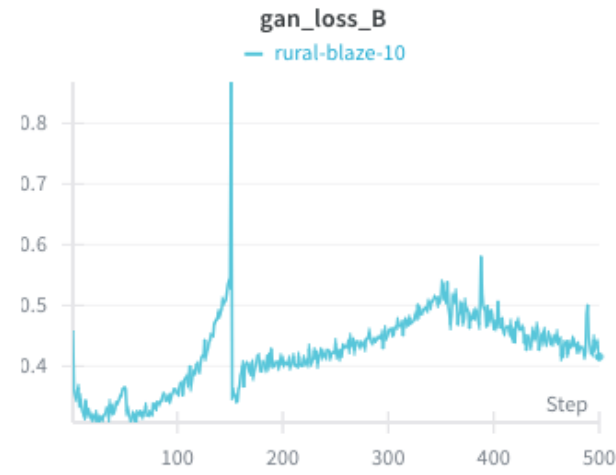


FIG. 14: GAN adversarial loss for domain B (gan_loss_B)

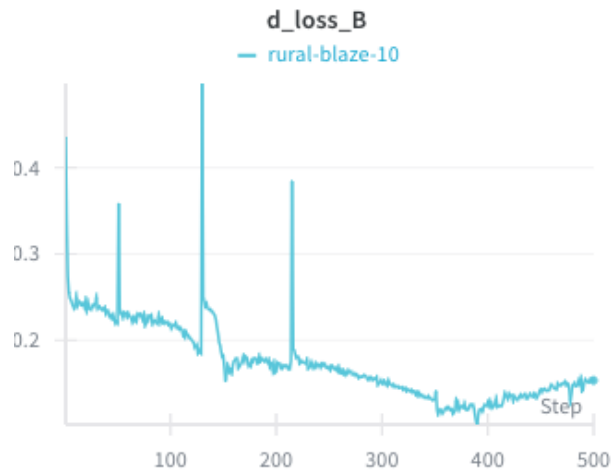


FIG. 16: Discriminator loss for domain B (d_loss_B)

4. Cycle Consistency and Structural Preservation Losses

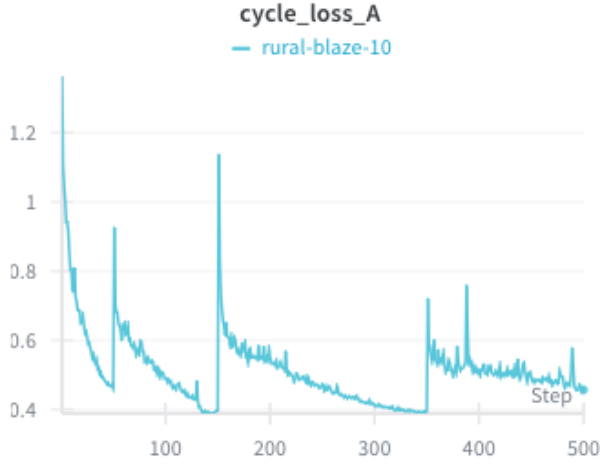


FIG. 17: Cycle consistency loss for domain A (cycle_loss_A)



FIG. 18: Cycle consistency loss for domain B (cycle_loss_B)

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